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Assessment of the impact of bubble size modelling in CFD simulations of alternative bubble column configurations operating in the heterogeneous regime

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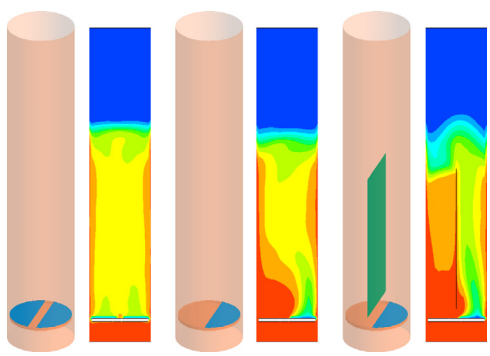
HIGHLIGHTS

- Effect of different bubble size modelling approaches evaluated.
- Variable single bubble size (VSBS) model gives reasonable agreement with data.
- VSBS model has the advantage of not needing bubble size as input.
- MUSIG model closures evaluated under-predict the bubble size.

GRAPHICAL ABSTRACT



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ABSTRACT

Modelling bubble columns operating in the churn turbulent flow regime is a considerable challenge due to their complex and transient behaviour. A key modelling decision is how best to model the bubble size distribution, with a range of approaches existing from the relatively simple option of using a single, representative bubble size to more complex population balance approaches. In this comparative study, we have examined various approaches (i.e. constant and variable single bubble size models, dual bubble size models, and both homogeneous and inhomogeneous MUSIG models). Model predictions were compared with experimental data from three different configurations of a pilot-scale column (0.39 m in diameter and 2 m in height) operated at industrially relevant superficial gas velocities of 0.16 and 0.25 m s^{-1}. The single bubble size models gave reasonable agreement with experimental data for the 'symmetrical' configuration, with the variable bubble size model having the major advantage of providing acceptable predictions of the mean bubble size (without this having to be provided as a model input). Implementation of multiple bubble size models did not lead to any substantial improvement in the modelling fit with the experimental data, leading to the inescapable conclusion that the standard bubble breakup and coalescence kernels proposed in the literature do not adequately describe bubble transformations taking place under heterogeneous operating conditions.

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1. Introduction

Bubble columns and airlift reactors are widely employed as gas-liquid contactors in both the chemical and biochemical industries.

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Nomenclature

AC	airlift column
BC	bubble column
BSD	bubble size distribution
C_1, C_2, C_3, C_μ	constants in k-ε model
C_4, C_5, C_6	constants in VSBS model
C_D	bubble drag coefficient
C_{LG}	momentum transfer coefficient for the interphase drag force [kg m^{-3} s^{-1}]
CFD	computational fluid dynamics
CSBS	constant single bubble size model
CTBS	constant two bubble size model
d_b	bubble diameter [m]
d_b^e	equilibrium bubble diameter in VSBS model [m]
D_{eff}	effective bubble dispersion coefficient [m^2 s^{-1}]
Eo	Eötvös number
f	size fraction
$f(\alpha_G)$	volume fraction correction term
$\boldsymbol{F_D}$	drag force acting on bubble [kg m^{-2} s^{-2}]
$\boldsymbol{F_{TD}}$	turbulence dispersion force per volume [kg m^{-2} s^{-2}]
$\boldsymbol{g}$	gravitational acceleration [m s^{-2}]
H, J	empirical function of the Morton and Eötvös dimensionless groups
HSC	half-sparger column
h-MUSIG	homogeneous MUSIG model
i-MUSIG	inhomogeneous MUSIG model
J	empirical function of the Morton and Eötvös dimensionless groups
k_L	liquid turbulence kinetic energy [m^2 s^{-2}]
M	number of size groups
$\boldsymbol{M}_{LG}$	gas-liquid interfacial forces [kg m^{-2} s^{-2}]
Mo	Morton number
MUSIG	multiple size group model
n	number of bubbles per unit volume [m^{-3}]
N	number of bubble velocity groups
p	pressure [Pa]
P_L	production of turbulence by shear [kg m^{-1} s^{-3}]
$Sc_{t,L}$	turbulent Schmidt number
S_{kj}	mass source term due to bubble breakup and coalescence [kg m^{-3} s^{-1}]
$\boldsymbol{S}_{M,j}$	momentum source due to bubble breakup and coalescence [kg m^{-2} s^{-2}]
St	Stokes number
t	time [s]
$T_{LG,\varepsilon}$	interphase turbulent energy dissipation transfer term [kg m^{-1} s^{-4}]
$T_{LG,k}$	interphase turbulent kinetic energy transfer term [kg m^{-1} s^{-3}]
$\boldsymbol{U}_{slip}$	bubble slip velocity [m s^{-1}]
$U_{b,\infty}$	bubble terminal velocity [m s^{-1}]
$\boldsymbol{U}_G$	gas velocity [m s^{-1}]
$\boldsymbol{U}_L$	liquid velocity [m s^{-1}]
VSBS	variable single bubble size model

Greek letters

α	volume fraction
$\boldsymbol{\delta}$	Kronecker delta function
ε	liquid turbulence energy dissipation rate [m^2 s^{-3}]
σ	surface tension [N m^{-1}]
σ_k	constant in k-ε model
σ_ε	constant in k-ε model
μ	dynamic viscosity [kg m^{-1} s^{-1}]
μ_{ref}	reference viscosity [kg m^{-1} s^{-1}]
$\mu_{t,L}$	turbulence viscosity [kg m^{-1} s^{-1}]
υ_L	kinematic viscosity [m^2 s^{-1}]
ρ	density [kg m^{-3}]
τ	dissipation timescale in Yao and Morel model [s]
τ_B	bubble breakup timescale in VSBS model [s]
τ_C	bubble coalescence timescale in VSBS model [s]
τ_L	characteristic time of the energetic turbulent eddies in VSBS model [s]
τ_p	bubble dynamic relaxation time in VSBS model [s]
τ_{rel}	relaxation time in VSBS model [s]
τ_t	turbulent microscale time in VSBS model [s]

Subscripts

b	bubble
D	drag force
eff	effective
G	gas phase
j	index for bubble phase
k	index for bubble sub-size class
L	liquid phase
LG	gas-liquid interphase
max	maximum
ref	reference
$slip$	slip velocity between gas and liquid
t	turbulence
TD	turbulent dispersion force

This is largely due to their mechanical simplicity, low capital cost, and good heat and mass transfer characteristics (Kantarci et al., 2005; Shah et al., 1982). Despite their apparent mechanical simplicity, the hydrodynamics occurring inside such columns is highly complex, particularly in the churn turbulent flow regime. Understanding and modelling their internal behavior is key to designing, operating and optimizing a wide range of industrial processes, and for this reason extensive use has been made of Computational Fluid Dynamics (CFD) (Jakobsen et al., 2005; Kelly, 2008; Rzehak et al., 2017; Sokolichin et al., 2004; Ziegenhein et al., 2017). In such CFD simulations, the Euler-Euler approach is typically used to model bubble columns operating at high superficial gas velocities (Becker et al., 1994; Bhole et al., 2008). Therefore, an accurate description of all key interactions, including interphase momentum transfer and turbulence sources, is central to the development of a meaningful process model. One fundamental process descriptor is the bubble size distribution (BSD). This is key as it determines the interfacial area for momentum, mass and heat transfer. A range of options exist here, from approaches that employ a single, constant bubble size to far more sophisticated population balance approaches that consider how bubble breakup and coalescence affect the way in which the BSD varies within the column (e.g. MUSIG (Lo, 1996) or QMOM (Marchisio et al., 2003)). Each approach has its advantages and limitations that manifest themselves in the trade-offs between accuracy (in terms of matching experimental data), model complexity, necessary modelling information and computational demand.

Some authors (McClure et al., 2014b; Rampure et al., 2007) have used a CFD model employing a constant, single bubble size (CSBS). Such a model has the advantages of simplicity and low computational demand (Chen et al., 2005b), while the central assumption is justified in situations where the bubble size distribution is inherently narrow (e.g. in fermentation media), or when the flow behavior can be reasonably approximated using a single bubble size. Pan

et al. (1999) described the flow regime within a two-dimensional bubble column using the CSBS model, with their model results showing good agreement with time-averaged experimental data. Similarly, our previous work (McClure et al., 2014a, 2014b) has demonstrated that it is possible to obtain acceptable agreement between experimental data and model predictions for a symmetrical bubble column configuration.

However, the choice of a constant bubble size throughout the whole column can never fully capture the reality of turbulent flow behavior where a wide spectrum of sizes may exist due to bubble breakup and coalescence under locally transient conditions. If the two-phase flow cannot be reasonably approximated using the mean bubble size (e.g. bubbles segregate based on size), it is necessary to use a more complex model which can account for the spatial and temporal variations in bubble size. Another obvious drawback of the CSBS model is that it necessitates *a priori* knowledge of the mean bubble size. In our previous studies, this has been obtained from detailed experimental measurements of the BSD; however, obtaining such data for industrial equipment is likely to be a considerable challenge.

Laux and Johansen (1999) presented an alternative approach to single bubble size models, referred to as the variable single bubble size (VSBS) model in this present study. This approach avoids the need to specify the bubble size in the simulation. Rather, the effect of bubble coalescence and breakup is incorporated into a transport equation for the average bubble size. All bubble sizes within a given control volume constitute a local size distribution that is determined by convective and diffusive fluxes together with a source term that accounts for bubble breakup and coalescence. One of the objectives of the present study was to evaluate the applicability of the VSBS model in simulating various bubble column configurations in terms of its ability to retain the simplicity of a single bubble size but incorporate (albeit indirectly) the effects of bubble breakup and coalescence.

Alternatively, to overcome the limitations of a constant single bubble size model, Krishna and van Baten (2001) developed a constant two bubble size (CTBS) model. In this case, the fluid in the bubble column is made up of three phases (i.e., the liquid, 'small' bubble and 'large' bubble phases) with the Eulerian description being used for each phase. Interactions between the liquid and the gas phases are accounted for in terms of momentum exchange, however interactions between the two gas phases are not considered. Their simulation results showed good agreement with experimental data on gas hold-up and liquid velocity profiles for a 0.38 m diameter column operated at 0.23 m s^{-1}, although it was appreciated that the need to specify both bubble sizes and the volume fraction associated with each gas phase would limit its *a priori* predictive ability.

A more complex approach to model the BSD is provided by the homogeneous MUSIG model proposed by Lo (1996). In this approach, the bubbles are divided into M size groups. Continuity equations are solved for each bubble size group to capture the local BSD. In this approach, all size groups share the same velocity field (i.e. only one set of momentum equations is solved for the gas phase). This modelling approach has led to some success in the prediction of bubbly flows. For example, Yuan et al. (2016) measured the BSD for both air-water and air-nanofluid systems in a vertical tube (0.015 m in diameter and 2.5 m in height), with the homogeneous MUSIG model providing satisfactory agreement for the air-water case but not for the air-nanofluid flow system.

However, it would seem that the application of this MUSIG model is best suited to homogeneous dispersed flows where the assumption that all bubbles share the same velocity field is a reasonable simplification. More recently, Frank et al. (2005) presented an inhomogeneous MUSIG model specifically for poly-dispersed flow systems. Here, the gas phase is divided into N velocity groups, with each group being characterized by its own velocity field. Furthermore, the complete BSD is represented by dividing the bubble size range into M groupings within each of the velocity groups with bubble coalescence and breakup accounted for across the entire bubble size range (Krepper et al., 2005). Extensive reviews of both bubble breakup and coalescence have been provided by Liao and Lucas (2009, 2010). Liao et al. (2011) proposed updated breakup and coalescence kernels for the inhomogeneous MUSIG model and compared their simulation results with experimental data obtained from the TOPFLOW test facility (0.2 m in diameter, gas velocities up to 0.09 m s^{-1}) with closer matches being obtained for the evolution of the BSD compared with simulated results that employed the earlier closure models of Luo and Svendsen (1996) and Prince and Blanch (1990). Despite these published successes, it is acknowledged that a key challenge in using both homogeneous and inhomogeneous MUSIG models as reliable predictive tools is the development of breakup and coalescence kernels that are applicable (and experimentally validated) across a wide range of operating conditions (Besagni et al., 2017; Rzehak et al., 2017).

In our previous work (Fletcher et al., 2017; McClure et al., 2017a, 2014b, 2016), extensive (and reproducible) experimental data were gathered from three different configurations (i.e. two bubble column configurations differing only in the sparger used, the third being a bubble column modified to operate as an airlift system), operated at relatively high superficial velocities (up to 0.37 m s^{-1}) typical of a range of industrial processes. Associated computational studies using a CSBS model gave acceptable agreement between simulated and experimental results for the bubble column, but agreement was less good for the asymmetric column configurations. Hence, the aim of this present study is to evaluate the applicability of different BSD modelling approaches to bubble columns and airlifts operated at high superficial velocities. As far as the authors are aware, this is the first published comparison of experimental data from closely related bubble column configurations operating in the heterogeneous regime (and covering both symmetrical and asymmetrical conditions) with CFD simulation results from a complete spectrum of bubble size modelling approaches, ranging from the simplest single bubble size to the full inhomogeneous MUSIG model.

2. Model description

2.1. Conservation equations

For cases where only one bubble size group is considered and there is no inter-phase mass transfer, the mass and momentum conservation equations for the gas and liquid phases are as follows:

$$\frac{\partial(\rho_G\alpha_G)}{\partial t} + \nabla\cdot(\rho_G\alpha_G\boldsymbol{U}_G) = 0 \tag{1}$$

$$\frac{\partial(\rho_L\alpha_L)}{\partial t} + \nabla\cdot(\rho_L\alpha_L\boldsymbol{U}_L) = 0 \tag{2}$$

$$\begin{aligned}\frac{\partial(\rho_G\alpha_G\boldsymbol{U}_G)}{\partial t} &+ \nabla\cdot(\rho_G\alpha_G\boldsymbol{U}_G\otimes\boldsymbol{U}_G)\\ &= -\alpha_G\nabla p + \nabla\cdot\left(\alpha_G\mu_{G,\mathrm{eff}}\left[\nabla\boldsymbol{U}_G + (\nabla\boldsymbol{U}_G)^T - \frac{2}{3}\boldsymbol{\delta}\nabla\cdot\boldsymbol{U}_G\right]\right)\\ &\quad + \alpha_G\rho_G\boldsymbol{g} + \boldsymbol{M}_{LG}\end{aligned} \tag{3}$$

$$\begin{aligned}\frac{\partial(\rho_L\alpha_L\boldsymbol{U}_L)}{\partial t} &+ \nabla\cdot(\rho_L\alpha_L\boldsymbol{U}_L\otimes\boldsymbol{U}_L)\\ &= -\alpha_L\nabla p + \nabla\cdot\left(\alpha_L\mu_{L,\mathrm{eff}}\left[\nabla\boldsymbol{U}_L + (\nabla\boldsymbol{U}_L)^T - \frac{2}{3}\boldsymbol{\delta}\nabla\cdot\boldsymbol{U}_L\right]\right)\\ &\quad + \alpha_L\rho_L\boldsymbol{g} - \boldsymbol{M}_{LG}\end{aligned} \tag{4}$$

where t is time, α is the volume fraction, ρ is the density, $\boldsymbol{U}$ is the velocity vector, p is the pressure, μ_{eff} is the effective viscosity, $\boldsymbol{g}$ is acceleration due to gravity, $\boldsymbol{M}_{LG}$ is the inter-phase momentum transfer term and δ is the Kronecker delta function. Here, the subscripts L and G refer to the liquid and gas phases, respectively.

In the MUSIG approach, several bubble size groups exist and the exchange of mass between them (caused by bubble breakup and coalescence) must be considered. The momentum equations for different velocity groups in the inhomogeneous MUSIG model also need to be considered. In the current work, two velocity groups ($N = 2$), where each bubble phase is further divided into three bubble sub-size groups ($M_j = 3$, $j = 1, 2$) were employed. The additional mass and momentum conservation equations for the two bubble phases in the inhomogeneous MUSIG model can be written as:

$$\frac{\partial(\rho_{G,j}\alpha_{G,j}f_k)}{\partial t} + \nabla\cdot(\rho_{G,j}\alpha_{G,j}\boldsymbol{U}_{G,j}f_k) = S_{kj} \tag{5}$$

$$\begin{aligned}\frac{\partial(\rho_{G,j}\alpha_{G,j}\boldsymbol{U}_{G,j})}{\partial t} &+ \nabla\cdot(\rho_{G,j}\alpha_{G,j}\boldsymbol{U}_{G,j}\otimes\boldsymbol{U}_{G,j}) \\ &= -\alpha_{G,j}\nabla p + \nabla\cdot\left(\alpha_{G,j}\mu_{G,\mathrm{eff}}\left[\nabla\boldsymbol{U}_{G,j} + (\nabla\boldsymbol{U}_{G,j})^T - \frac{2}{3}\delta\nabla\cdot\boldsymbol{U}_{G,j}\right]\right) \\ &\quad + \alpha_{G,j}\rho_{G,j}\boldsymbol{g} + \boldsymbol{M}_{LG,j} + \boldsymbol{S}_{M,j}\end{aligned} \tag{6}$$

where the subscript $j = 1, 2$ represent the large and small bubble phases, and the subscript $k = 1, 2, 3$ represent the bubble sub-size groups, respectively. f_k is defined as the size fraction of the sub-size group k, and satisfies the relationship $\sum_{k=1}^{M_j} f_k = 1$. The terms S_{kj} and $\boldsymbol{S}_{M,j}$ are source terms for mass and momentum transfer between velocity groups due to bubble breakup and coalescence processes.

In the variable single bubble size model, the mean diameter of the local bubble size distribution is described by a transport equation using the model of Laux and Johansen (1999):

$$\frac{\partial(\rho_G\alpha_G d_b)}{\partial t} + \nabla\cdot(\rho_G\alpha_G(d_b\boldsymbol{U}_G - D_{\mathrm{eff}}\nabla d_b)) = \rho_G\alpha_G\frac{(d_b^e - d_b)}{\tau_{rel}} \tag{7}$$

where d_b is the instantaneous mean bubble diameter and D_{eff} is an effective bubble dispersion coefficient which for turbulent flow is given by:

$$D_{\mathrm{eff}} = \frac{\mu_{t,L}}{\rho_L} \tag{8}$$

The term on the right-hand side of Eq. (7) drives the mean bubble diameter towards an equilibrium value, d_b^e, which is determined by the local hydrodynamic conditions (Calderbank, 1958):

$$d_b^e = C_4\alpha_G^{C_6}\frac{\left(\frac{\sigma}{\rho_L}\right)^{0.6}}{\epsilon_L^{0.4}} + C_5 \tag{9}$$

where the constants C_4, C_5 and C_6 take the values of 4.15, 0.9 and 0.5, respectively.

Here, τ_{rel} is the characteristic time for the mean bubble size to reach its equilibrium value, which is related to the timescales for breakup (τ_B), coalescence (τ_C) and the turbulent microscale (τ_t) (Laux and Johansen, 1999):

$$\tau_{rel} = \max(\tau'_{rel}, \tau_t) \tag{10}$$

$$\tau'_{rel} = \begin{cases}\tau_B & d_b > d_b^e \\ \tau_C & d_b < d_b^e\end{cases} \tag{11}$$

$$\tau_t = 6\sqrt{\frac{\upsilon_L}{\varepsilon}} \tag{12}$$

where υ_L is the kinematic viscosity of the liquid. The breakup timescale (τ_B) is based on the shearing force caused by turbulent eddies of a similar size to the bubble diameter and is determined using the length-scale from the bubble diameter and the turbulence energy dissipation rate, as given in Eq. (32).

Laux and Johansen (1999) assumed that coalescing bubbles are brought into contact by turbulent velocity fluctuations, and modelled the time scale for coalescence (τ_C) accordingly:

$$\tau_C = 2\left[\frac{\pi}{6}\frac{(\alpha_{G,\max} - \alpha_G)}{\alpha_G}\right]^{\frac{1}{3}}\frac{d_b}{\sqrt{\frac{2}{3}k_L}}\sqrt{1 + \mathrm{St}} \tag{13}$$

where $\alpha_{G,\max}$ is the maximum gas volume fraction (a value of 1 was used in the present numerical study) and St is the Stokes number for the bubbles, related to the bubble relaxation time (τ_p) and the timescale for turbulent eddies (τ_L) as follows:

$$\mathrm{St} = \frac{\tau_p}{\tau_L} \tag{14}$$

$$\tau_p = \frac{\rho_G d_b^2}{18\mu_L} \tag{15}$$

$$\tau_L = \frac{1}{n\pi d_b^2|\boldsymbol{U}_{slip}|} \tag{16}$$

where the bubble slip velocity ($\boldsymbol{U}_{slip}$) was calculated using the instantaneous values of the gas and liquid phase velocities.

Here, we have used both the homogeneous and inhomogeneous MUSIG models as implemented in ANSYS CFX for which the conservation equations and closure relationships are fully described in Liao et al. (2011) and only the key points are summarised here. The model of Luo and Svendsen (1996) was used for bubble breakup while the model of Prince and Blanch (1990) was used to describe coalescence. Luo and Svendsen employed a probabilistic approach with bubble breakup due to isotopic turbulence. Their model was designed and validated for air-water bubbly flow in high intensity pipeline flow. The pipe diameters used were small (25–50 mm) with reported bubble sizes of 1.5–4.3 mm and calculated turbulence energy dissipation rates of 5–70 W kg^{-1}. Although such a breakage model is undoubtedly best suited to the homogeneous operating regime, their bubble size and turbulence energy dissipation rate data are not dissimilar to those encountered in our experimental system, and so in the absence of any better validated description of bubble breakage, Luo and Svendsen's model was employed in our computational work. The coalescence model of Prince and Blanch (1990) is based on bubble collision due to turbulence, buoyancy and laminar shear effects, resulting in the formation of a thin liquid film that drains and then ruptures at a critical thickness. This model was validated against data from a bubble column with a diameter of 0.27 m, a water depth of 2 m, and gas superficial velocities up to 0.05 m s^{-1}. Such validation data are clearly most relevant to columns operating in the homogeneous regime, but again due to the absence of any more appropriate model, this closure relation was adopted in our MUSIG calculations.

2.2. Turbulence model

There is a significant literature on turbulence modelling in bubble columns, which is particularly well summarised in Khan et al. (2017). This recent review shows that the k-ε model (or Shear Stress Transport (SST) model which will be operating in k-ε mode on the meshes used) is overwhelmingly the most widely used. Reynolds stress turbulence models have also been employed by several authors, with Liu and Hinrichsen (2014) comparing the k-ε description with the computationally more demanding Reynolds stress model in the heterogeneous regime. These authors found lit-

tle benefit to using the Reynolds stress model and, in fact, their results showed that the closure modes for bubble drag and lift each played a much more significant role in the predicted results than the choice of turbulence model. Large Eddy Simulations (LES) or Scale Resolving Simulations (SRS) which seek to resolve a significant fraction of the turbulence spectrum have also been employed (e.g. Deen et al., 2001; Ničeno et al., 2008; Ma et al., 2015; Liu and Li, 2018) but these are more suitable for a detailed investigation of the fundamental physics than for the development of an 'industrial strength' model. They tend to be restricted to columns operating in the homogeneous regime and require fine computational meshes. Indeed, recent work by Liu and Li (2018) showed that mesh sizes smaller than the actual bubbles are needed to obtain reasonable mesh independence, something which is hard to reconcile with the use of phase-averaged equations. Bubble column modelling studies, including our own (McClure et al., 2014a), have clearly demonstrated that the inclusion of Bubble Induced Turbulence (BIT) is crucial. Such models have been formulated and computationally explored far more widely within the k-ε turbulence modelling framework (see, for example, Rzehak and Krepper, 2013) which is yet another reason that the two-equation turbulence modelling approach is often the preferred option.

Therefore, in light of the previous discussion, in this present study turbulence in the liquid phase was modelled using the standard k-ε model, whilst the dispersed phase zero equation model was used for the gas phase. The transport equations for the turbulence kinetic energy (k_L) and the dissipation rate of turbulence kinetic energy (ε_L) for the liquid phase are thus:

$$\frac{\partial(\rho_L\alpha_L k_L)}{\partial t} + \nabla\cdot\left(\alpha_L\left(\rho_L k_L \boldsymbol{U}_L - \left(\mu_L + \frac{\mu_{t,L}}{\sigma_k}\right)\nabla k_L\right)\right) = \alpha_L(P_L - \rho_L\varepsilon_L) + T_{LG,k} \tag{17}$$

and

$$\frac{\partial(\rho_L\alpha_L \varepsilon_L)}{\partial t} + \nabla\cdot\left(\alpha_L\left(\rho_L \varepsilon_L \boldsymbol{U}_L - \left(\mu_L + \frac{\mu_{t,L}}{\sigma_\varepsilon}\right)\nabla \varepsilon_L\right)\right) = \alpha_L\frac{\varepsilon_L}{k_L}(C_1P_L - C_2\rho_L\varepsilon_L) + T_{LG,\varepsilon} \tag{18}$$

where P_L is the turbulence production due to shear. The constants C_1, C_2, σ_k and σ_ε take the standard values of 1.44, 1.92, 1.0 and 1.3, respectively (ANSYS, 2017). $T_{LG,k}$ and $T_{LG,\varepsilon}$ are source terms related to bubble-induced turbulence whilst the liquid phase turbulent viscosity ($\mu_{t,L}$) is modelled as:

$$\mu_{t,L} = C_\mu\rho_L\left(\frac{k_L^2}{\varepsilon_L}\right) \tag{19}$$

where C_μ was set to 0.09 and no bubble-induced turbulence was included as part of this viscosity term.

2.3. Inter-phase transfer terms

Following McClure et al. (2014b), only interphase drag and turbulence dispersion have been modelled in the present work, so that $\boldsymbol{M}_{GL} = \boldsymbol{F_D} + \boldsymbol{F_{TD}}$. Here, both the lift force and virtual mass forces are not included as they have previously been shown to have only a small effect in symmetrical column simulations and may reduce model stability markedly (Tabib et al., 2008). The drag force per volume ($\boldsymbol{F}_D$) for spherical bubbles is given by:

$$\boldsymbol{F}_D = \frac{3}{4}\frac{C_D}{d_b}\rho_L\alpha_G f(\alpha_G)(\boldsymbol{U}_G - \boldsymbol{U}_L)|\boldsymbol{U}_G - \boldsymbol{U}_L| \tag{20}$$

where C_D is the drag coefficient, d_b is the bubble diameter, and $f(\alpha_G)$ is the volume fraction correction term.

The drag force experienced by an isolated bubble was determined using the correlation developed by Clift et al. (1978), where the drag coefficient is taken from the terminal velocity of an isolated bubble ($U_{b,\infty}$) as defined by:

$$U_{b,\infty} = \frac{\mu_L}{d_b\rho_L}\mathrm{Mo}^{-0.149}(J - 0.857), \quad \mathrm{Mo} < 0.001 \tag{21}$$

where μ_L is the liquid viscosity and J and H are empirical functions of the Morton (Mo) and Eötvös (Eo) dimensionless groups:

$$J = \begin{cases} 0.94H^{0.757}, & 2 < H \leqslant 59.3 \\ 3.42H^{0.441}, & H > 59.3 \end{cases} \tag{22}$$

$$H = \frac{4}{3}\mathrm{EoMo}^{-0.149}\left(\frac{\mu_L}{\mu_{ref}}\right)^{-0.14}, \quad \mathrm{Eo} < 40 \tag{23}$$

μ_{ref} is the reference viscosity (9×10^{-4} Pa·s); whilst the Morton (Mo) and Eötvös (Eo) numbers are defined as follows:

$$\mathrm{Mo} = \frac{\mu_L^4 g(\rho_L - \rho_G)}{\rho_L^2\sigma^3} \tag{24}$$

$$\mathrm{Eo} = \frac{g(\rho_L - \rho_G)d_b^2}{\sigma} \tag{25}$$

where σ is the air-water interfacial tension (0.072 N/m). The drag coefficient (C_D) is related to the bubble terminal velocity as follows:

$$C_D = \frac{4gd_b(\rho_L - \rho_G)}{3U_{b,\infty}^2\rho_L} \tag{26}$$

To account for bubble drag reduction due to a high local gas volume fraction, a modified form (Fletcher et al., 2017) of the Simonnet et al. (2007) correction term $f(\alpha_G)$ was used in this study:

$$f(\alpha_G) = \begin{Bmatrix} \min(f'(\alpha_G), 1.0) & f'(\alpha_G) > 1 \\ 0.8f'(\alpha_G) & f'(\alpha_G) < 1 \end{Bmatrix} \tag{27}$$

$$f'(\alpha_G) = (1 - \alpha_G)\times\left[(1-\alpha_G)^{25} + \left(4.8\frac{\alpha_G}{1-\alpha_G}\right)^{25}\right]^{-\frac{2}{25}} \tag{28}$$

Turbulent dispersion was also included, modelled using the Favre-averaged drag relationship developed by Burns et al. (2004):

$$\boldsymbol{F}_{TD} = C_{LG}\frac{\upsilon_L}{Sc_{t,L}}\left(\frac{\nabla\alpha_G}{\alpha_G} - \frac{\nabla\alpha_L}{\alpha_L}\right) \tag{29}$$

where C_{LG} is the interphase momentum transfer coefficient due to drag and $Sc_{t,L}$ is the turbulent Schmidt number for the continuous phase which has a default value of 0.9.

The model developed by Yao and Morel (2004) was used for turbulence production due to the relative motion of the bubbles. In this model:

$$T_{LG,k} = \alpha_L C_1|\boldsymbol{M}_{LG}||\boldsymbol{U}_G - \boldsymbol{U}_L| \tag{30}$$

$$T_{LG,\varepsilon} = \frac{C_3\alpha_L C_1}{\tau}|\boldsymbol{M}_{LG}||\boldsymbol{U}_G - \boldsymbol{U}_L| \tag{31}$$

where C_3 was set to a value of 1.0. In this model, the dissipation timescale τ is determined using a relative length-scale between the bubble diameter (d_b) and ε:

$$\tau = \left(\frac{d_b^2}{\varepsilon}\right)^{\frac{1}{3}} \tag{32}$$

3. Model setup

In this study, CFD simulations were performed for three pilot-scale experimental set-ups as described by McClure et al. (2017a, 2014b, 2016). All three configurations have the same column

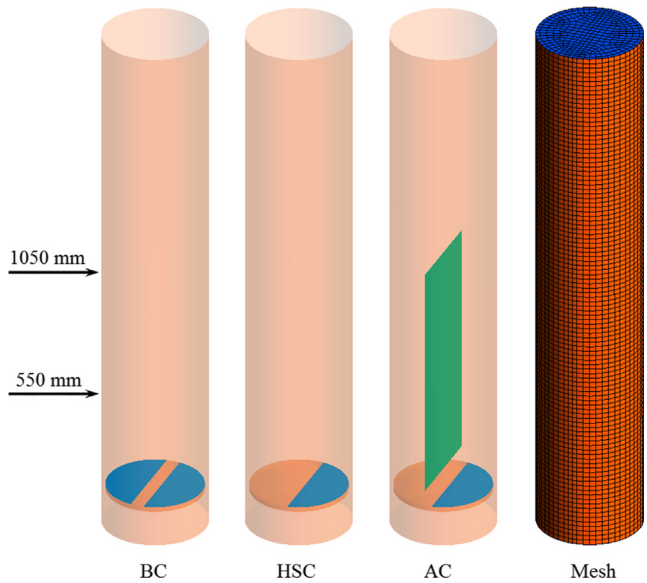

Fig. 1. Schematic showing column geometry and the computational mesh used for the three simulations.

diameter (0.39 m) and height (2 m), with the differences related to the sparger design and choice of column internals. As shown in Fig. 1, the three configurations were a bubble column with a symmetrical sparger (referred to as BC); a bubble column where the air was introduced asymmetrically (using a 'half-sparger' – HSC) and an airlift (AC) where a baffle was located on the column centerline. The initial water depth was 1.08 m and the initial void fraction in the model was set to 10^{-6} to avoid numerical issues at start-up.

In the simulations, bubbles were introduced at two semicircular areas corresponding to the stems of the 'tree' type sparger used for the bubble column, while the air was introduced on one side of the column for both the half-sparger and airlift configurations, as illustrated by the blue[1] shaded areas in Fig. 1. Superficial velocities of 0.16 and 0.25 m s^{-1} were used for all three cases, these being typical values for high intensity industrial operation. The top surface in each case was modelled as an opening boundary condition; walls were treated as no-slip boundaries for the liquid and free-slip boundaries for the gas phase. On the basis of previous work (McClure et al., 2014b), the same computational mesh, comprising 36,000 hexahedral cells, was used for all three column configurations. The computational mesh employed is also shown in Fig. 1.

For both the constant and variable single bubble size models, bubbles were introduced having a size of 8 mm for the bubble column and half-sparger configurations and 7 mm for the airlift; these values correspond to experimentally measured (McClure et al., 2017a, 2014b, 2016) mean bubble diameters.

For the homogeneous MUSIG model, bubbles from 1 to 24 mm diameter were divided into 6 size groups, with all groups sharing the same velocity field. These size groups were subdivided into two velocity groups for the inhomogeneous MUSIG model. The bubble size and velocity groups employed in the MUSIG models are summarised in Table 1. For both MUSIG approaches, bubbles were introduced into the column in size group 4 (i.e. 8 mm in diameter).

[1] For interpretation of color in Fig. 1, the reader is referred to the web version of this article.

Table 1
Bubble size and velocity groups used in the MUSIG models.

h-MUSIG model			*i*-MUSIG model		
Size group	Size (mm)	Velocity group	Size group	Size (mm)	Velocity group
1	1	One group	1	1	'small' bubbles
2	3		2	3	
3	5		3	5	
4	8		4	8	'large' bubbles
5	16		5	16	
6	24		6	24	

Table 2
Summary of the relative simulation timings and key characteristics of various bubble size models.

Bubble size model	Relative timing	Key advantages	Key disadvantages
CSBS	1.00	Simple; Computationally efficient	Need to determine the bubble size; No information provided on BSD
VSBS	1.06	Simple; Computationally efficient; No need to specify the bubble size	Limited information provided on BSD
CTBS	1.76	Simple	Increased computational demand; Need to define *a priori* both the size and volume fraction for each group
h-MUSIG	1.25	Provides more information on the BSD	Only one velocity field; Increased computational demand; Need to specify the number of size groups; Need for appropriate (and validated) breakup and coalescence kernels
i-MUSIG	2.06	Conceptually the most realistic; Provides detailed information on the BSD	Increased computational demand; Need to specify the number of velocity and size groups; Need for appropriate (and validated) breakup and coalescence kernels

All CFD models employed were implemented using a commercial software package (ANSYS CFX 18.2) with a three-dimensional Euler-Euler approach used. All cases were modelled as transient two-phase flow systems with a second-order backward Euler transient scheme employed. A high-resolution scheme was used for all convective terms, except for the turbulence equations where a first-order scheme was applied.

For reasons of numerical stability, up to 200 time-steps were initially performed using the segregated volume fraction solver with all subsequent time-steps using a coupled volume fraction solver. In all cases, simulations were carried out for a duration of 120 s of real-time, with a time step of 0.001 s being used. To prevent startup bias, the first 20 s of all simulation results were discarded before time-averaging was started. Unless stated otherwise, all reported results are time-averaged values. Relative computational times for each model (shown in Table 2) were calculated by simulating 2 s of column operation on six computational cores using the Artemis HPC cluster at the University of Sydney. For purposes of comparison, simulation of 2 s of column operation took approximately 2 h using the CSBS model.

4. Results and discussion

4.1. Single bubble size models

Comparisons of local volume fraction and upwards liquid velocity profiles against experimental measurements are presented in Figs. 2–4 for a superficial gas velocity of 0.16 m s^{-1}. Analogous results for a superficial velocity of 0.25 m s^{-1} show very similar trends, and thus are not included here.

Agreement between CFD predictions and experimental results is reasonable for the bubble column configuration using the CSBS model, as shown in Fig. 2. However, when the asymmetric half-sparger was used, there are quite noticeable differences between experimental and simulated results, as shown in Fig. 3.

Fig. 4 shows that agreement between predicted and experimental results for the local gas volume fraction is generally reasonable in the riser section for the airlift configuration, but significant differences are obvious in the downcomer section. Interestingly, the model under-predicts the amount of gas that is entrained in the downwards flowing liquid, see Fig. 4(a), while it over-predicts the gas fraction at the top area of the baffle in the downcomer section, see Fig. 4(b). On the other hand, the CSBS model over-predicts the liquid velocity in the riser section, while providing acceptable agreement in the downcomer region, as shown in Fig. 4(c) and (d).

Figs. 2–4 show that the gas volume fraction and upwards liquid velocity profiles predicted using the VSBS model are very similar with those from the CSBS model for all three configurations. Similar prediction trends are seen in the contour plots of the time-averaged gas volume fraction and liquid turbulence eddy dissipation rate, as illustrated in Figs. 5 and 7, respectively.

Contour plots of average bubble diameter for the three cases shown in Fig. 6 reveal that the predicted average bubble sizes using the VSBS model are principally in the range 7–8 mm for the bubble column and 6–8 mm for the half-sparger and airlift configurations. This agrees well with the experimental mean values (8 mm for the bubble column and half-sparger column; 7 mm for the airlift column). Simulations were also performed using initial bubble sizes both significantly below (2 mm) and above (25 mm) the experimentally measured mean value to determine the sensitivity of the VSBS model to the inlet conditions. These results are not shown but in all cases varying the inlet

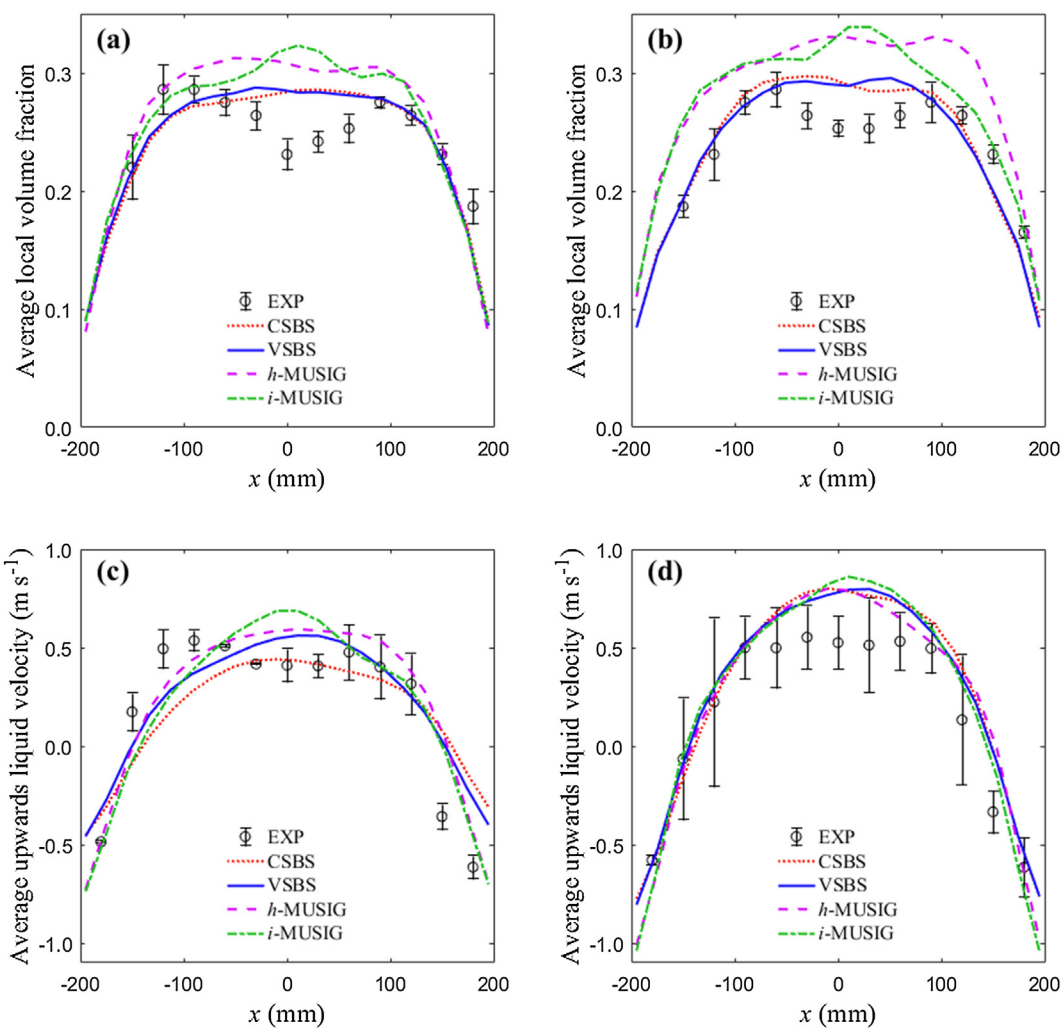

Fig. 2. Plots showing the local volume fraction and average upwards liquid velocity for the bubble column configuration for a superficial gas velocity of 0.16 m s^{-1}. The top row shows hold-up profiles at heights of (a) 550 mm and (b) 1050 mm; while the bottom row shows profiles of the average upwards liquid velocity at (c) 550 mm and (d) 1050 mm above the base of the column.

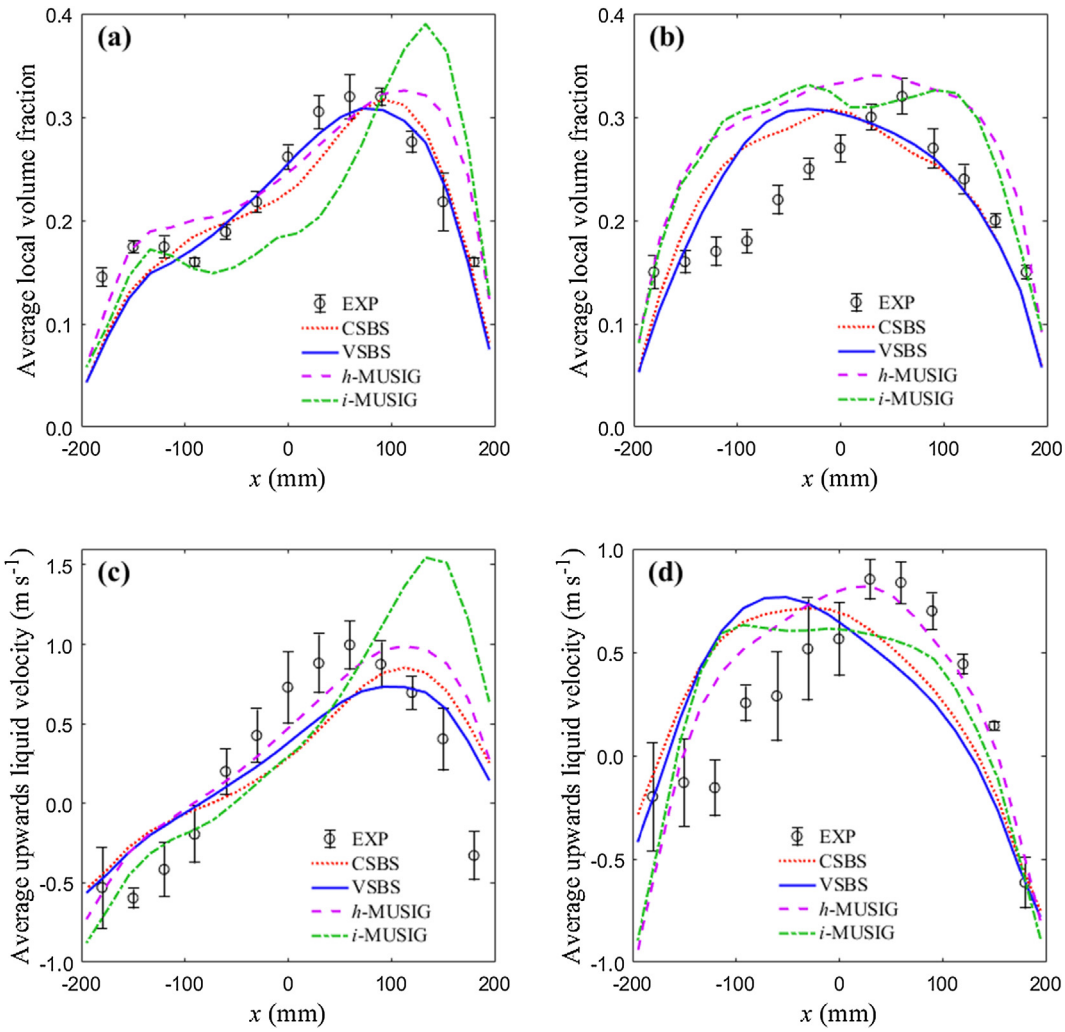

Fig. 3. Plots showing the local volume fraction and average upwards liquid velocity for the half-sparger column configuration for a superficial gas velocity of 0.16 m s^{-1}. The top row shows hold-up profiles at heights of (a) 550 mm and (b) 1050 mm; while the bottom row shows profiles of the average upwards liquid velocity at (c) 550 mm and (d) 1050 mm above the base of the column.

size had minimal impact on the model predictions, with bubbles quickly reaching their equilibrium size. Such results are encouraging as they indicate that VSBS model predictions are quite insensitive to the choice of inlet boundary condition and that the predicted equilibrium size is in good agreement with previously determined experimental data.

Table 2 summarises the relative 'computational cost' for each bubble size model. It is clear that employing the VSBS model adds minimal computational cost relative to the CSBS model. Such results suggest that the VSBS model could be useful in cases where the mean bubble size is not known *a priori* which can often be the case in industrial situations where it is difficult to make detailed experimental measurements. However, it is equally true that implementation of the VSBS model did not lead to improved model predictions for the two asymmetric configurations considered; an observation that is best attributed to the fact that having a single bubble size is not capturing all key aspects of the physical behavior occurring within the column. At this stage in the numerical study, it was felt that the more complex MUSIG models (albeit with their previously acknowledged limitations) should be assessed as to their ability to describe our asymmetric column configurations.

4.2. MUSIG models

Predicted profiles of the average gas hold-up and liquid velocity obtained using both homogeneous and inhomogeneous MUSIG models for the three column configurations are also given in Figs. 2–4. Both MUSIG models resulted in an increase in the gas volume fraction throughout the column, especially towards the top of the column. This tendency was confirmed via contour plots of the average gas volume fraction, as shown in Fig. 5, where one can clearly see that the predicted gas hold-up using the two MUSIG models is higher than those for the CSBS and VSBS models.

To better understand the MUSIG model predictions, time-averaged contour plots for the mean bubble size are given in Fig. 6. Somewhat surprisingly, bubble sizes obtained using both the homogeneous and inhomogeneous MUSIG models are generally below 3 mm for all three column configurations. In other words, bubbles entering the column (having a diameter of 8 mm) rapidly disappear by breakage to form smaller bubbles that undergo very limited coalescence.

These results suggest that the combination of the breakup model of Luo and Svendsen (1996) and the coalescence model of Prince and Blanch (1990), as implemented in the ANSYS CFX soft-

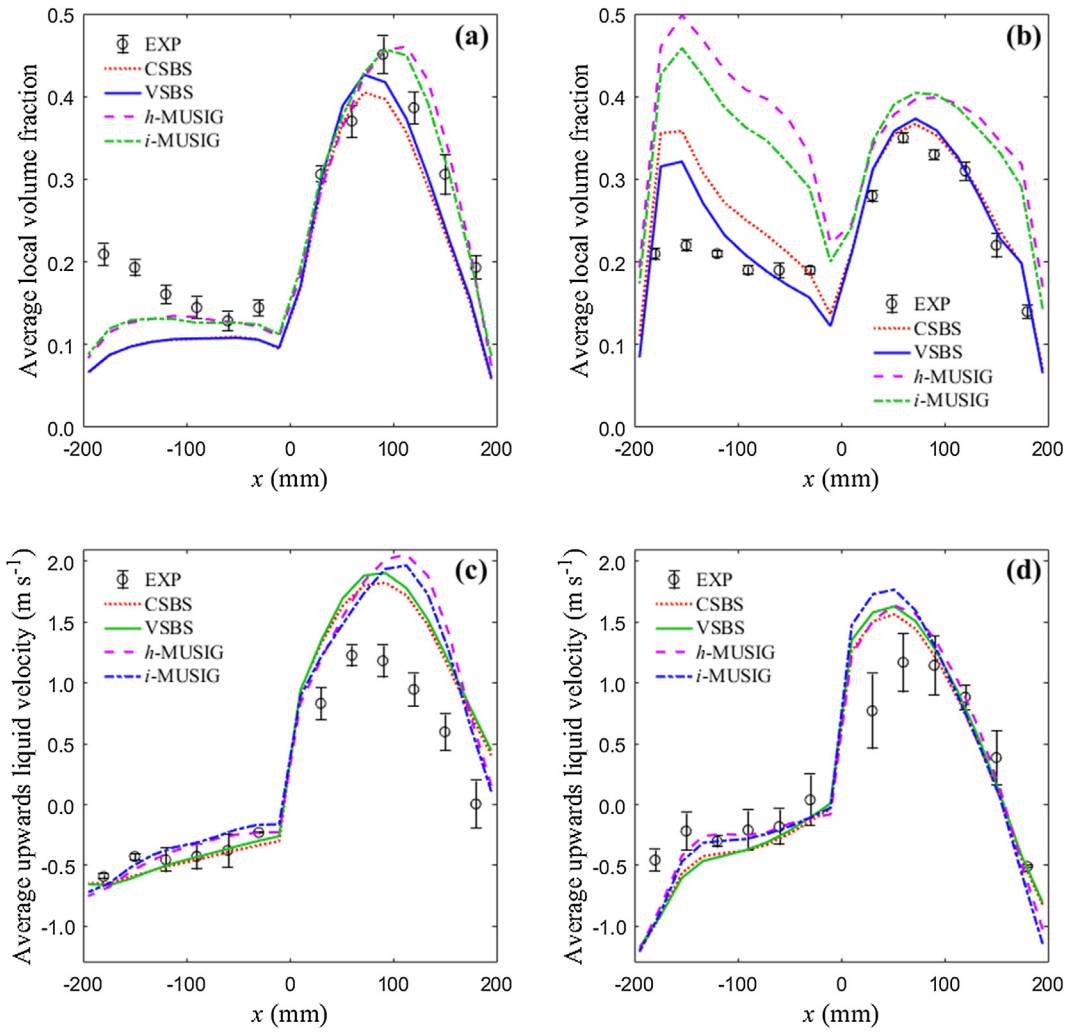

Fig. 4. Plots showing the local volume fraction and average upwards liquid velocity for the airlift column configuration for a superficial gas velocity of 0.16 m s^{-1}. The top row shows hold-up profiles at heights of (a) 550 mm and (b) 1050 mm; while the bottom row shows profiles of the average upwards liquid velocity at (c) 550 mm and (d) 1050 mm above the base of the column.

ware, results in an over-prediction of the number of small bubbles under high superficial velocity operating condition; a finding that agrees with those of Liao et al. (2011). One reason for this significant mismatch between our experimental and simulated results could be that the breakup model of Luo and Svendsen (1996) does not consider any force balance constraint but only takes into account the energy constraint (Lehr et al., 2002; Wang et al., 2003), which may lead to an over-prediction of the bubble breakup rate. However, it is equally possible that our observed discrepancy is due to an under-prediction of the coalescence rate. At this stage, without detailed experimental work to independently measure coalescence and breakup rates under heterogeneous conditions, it is not possible to identify whether the under-prediction of the mean bubble size is caused by excessive breakup and/or insufficient coalescence within these MUSIG models.

Interestingly, the over-estimation of small bubbles using MUSIG models in the present study is not consistent with published results by Chen et al. (2005a, 2004) who examined the effect of various closure options for both bubble breakup and coalescence, and concluded that the type of kernels used did not have a great effect on the predicted hold-up and velocity field. They also found that the model-predicted bubble coalescence rate was larger than the breakup rate by about an order of magnitude. Therefore, they arbitrarily increased the breakup rate by a factor of 10 to match their experimental data. Olmos et al. (2001) had previously used a MUSIG model for Euler-Euler simulations of a cylindrical bubble column (0.1 m in diameter and 1.35 m in height), and found that the Luo and Svendsen (1996) and the Prince and Blanch (1990) models each predicted too much breakup and coalescence, respectively. Thus, they arbitrarily modified both the bubble breakup and coalescence rates by a factor of 0.075 in their simulations to fit their experimental data. Similarly, Liao et al. (2011) developed alternative closure models, finding these gave better agreement with their experimental data than the Luo and Svendsen (1996) breakup and Prince and Blanch (1990) coalescence models. Overall, there is little doubt that models based on multiple bubble sizes have the potential to provide improved agreement with experimental data, although there is a clear need for improved closures that describe bubble coalescence and breakup under a wider range of operating conditions and to have these implemented as robust options within commercial CFD software.

4.3. Constant two bubble size model

To complete the present numerical study, simulations were also performed using the constant two bubble size (CTBS) model for the

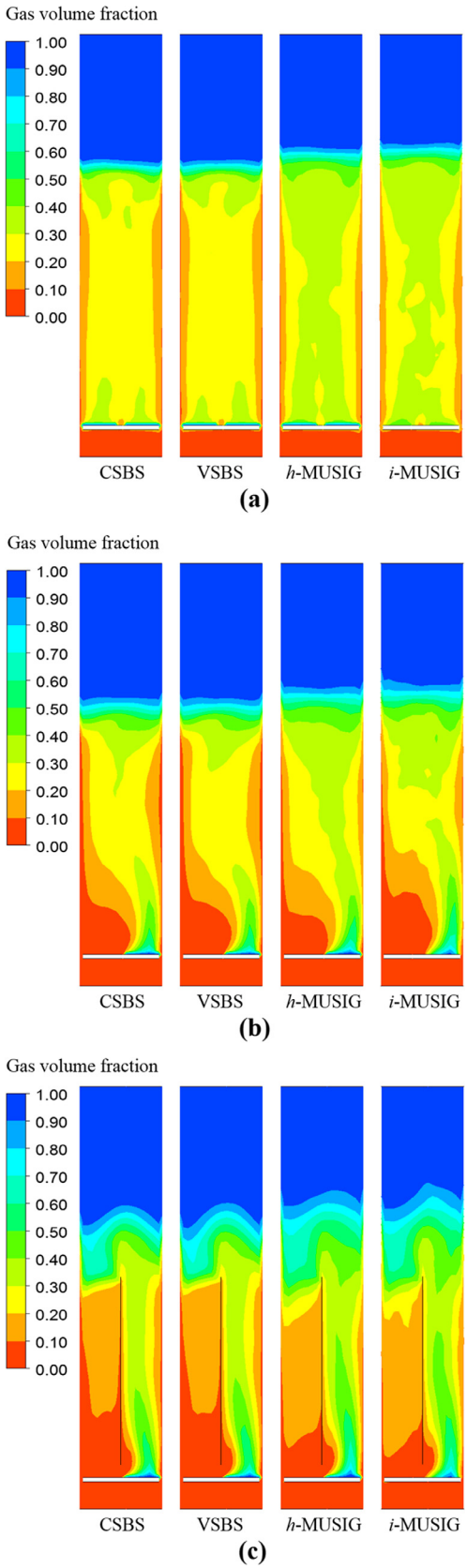

Fig. 5. Comparison of the predicted average local volume fraction for various bubble size models for the (a) bubble column, (b) half-sparger column and (c) airlift column configurations. The results shown here are for a superficial gas velocity of 0.16 m s^{-1}.

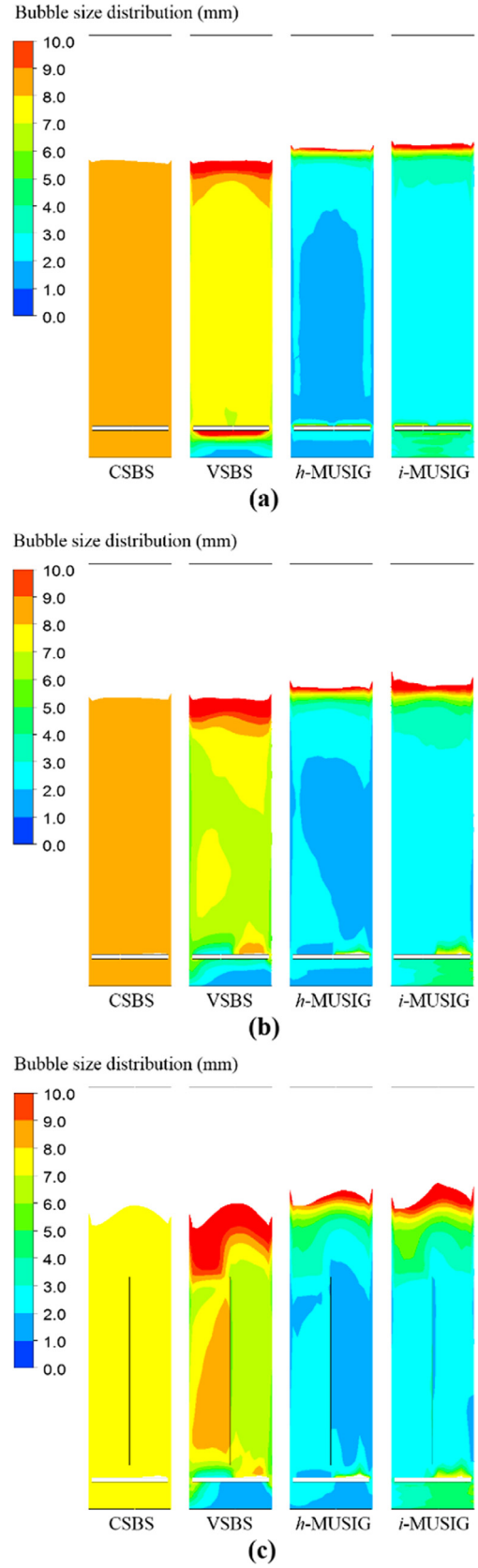

Fig. 6. Comparison of the predicted average bubble size distribution for various bubble size models for the (a) bubble column, (b) half-sparger column and (c) airlift column configurations. The results shown here are for a superficial gas velocity of 0.16 m s^{-1} and only for regions where the average gas volume fraction is below 0.9. For the airlift cases, air was introduced on the right with the baffle located on the column centerline. For comparative purposes the results from the CSBS model have also been shown; with the plotted bubble size corresponding to the experimental mean.

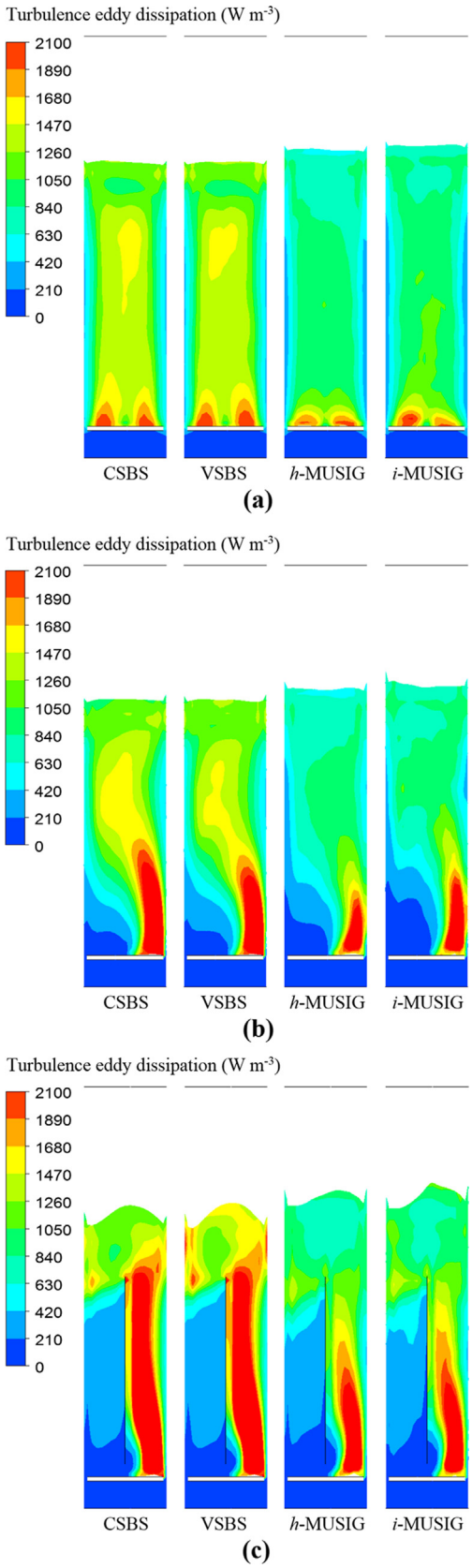

Fig. 7. Comparison of the predicted average liquid turbulence energy dissipation rate for various bubble size models for the (a) bubble column, (b) half-sparger column and (c) airlift column configurations. The results here are for a superficial gas velocity of 0.16 m s^{-1} and are only shown for regions having an average gas volume fraction below 0.9.

airlift configuration. In this model, the gas phase was divided into two size groups: 'small' and 'large' bubbles with no coalescence or breakup occurring for either. This (somewhat simplistic) model was designed to determine whether using a multi-size approach could give better agreement with experimental data while avoiding the issues around the selection of appropriate models for bubble breakup and coalescence.

Based on the bubble size distribution measured by McClure et al. (2017a), 3 mm and 12 mm bubbles were used here. These small and large bubbles were introduced into the column through the gas sparger in three different proportions: i.e. CTBS-1 (80% small bubbles; 20% large bubbles), CTBS-2 (50% small bubbles; 50% large bubbles), and CTBS-3 (20% small bubbles; 80% large bubbles) with all percentages being on a volume basis.

Comparisons between the experimentally measured and predicted profiles for the average hold-up and upwards liquid velocity are presented in Fig. 8. Interestingly, all CTBS models offered comparable (and reasonable) predictions of the average upwards liquid velocity. The major difference between the models was the extent to which smaller bubbles were entrained in the downcomer, as shown in Fig. 8(b). At the top of the downcomer (i.e. at a height of 1050 mm), the gas volume fraction increases as the fraction of small bubbles increases, a finding in line with the previous predictions of the MUSIG models, as well as what would be expected. However, none of the CTBS models correctly predicted the extent to which bubbles are entrained in the downcomer, particularly near the wall (as shown in Figs. 4(a) and 8(a)). Fig. 9 compares the predicted average local gas volume fraction, mean bubble size distribution and liquid turbulence eddy dissipation rate calculated using the CTBS models. Interestingly, changes in the fractions of small and large bubbles had little impact on the time-averaged gas volume fraction and liquid turbulence energy dissipation rate, as shown in Fig. 9(a) and (c).

Overall, bubble movement within the airlift configuration was very much as expected. However, for the CTBS model to be capable of providing closer agreement with experimental measurements would necessitate, at the very least, implementing validated size-dependent interfacial closures. For example, there is evidence that the magnitude of the volume fraction correction term depends on the bubble size (McClure et al., 2017b; Rabha and Buwa, 2010), while other inter-phase momentum transfer mechanisms (i.e. lift) are known to be bubble size dependent (Tomiyama et al., 2002).

4.4. Summary

A summary of the key advantages and disadvantages of the different models employed in this numerical study, along with their relative computational demands, is presented in Table 2. While it is accepted that the use of single bubble size models is undoubtedly a simplification, they are computationally efficient and offer surprisingly good agreement with available experimental data for most cases examined. Employment of the variable single bubble size (VSBS) model is particularly advantageous as it also gives a reasonable prediction of the mean bubble size, avoiding the need to have *a priori* knowledge of the most appropriate single bubble size. This would be especially beneficial for the predictive modelling of industrial systems where performing BSD measurements can be challenging and the presence of surface active materials can mitigate against a broad BSD.

In both homogeneous and inhomogeneous MUSIG models, the bubble breakup and coalescence kernel functions are central to the predictions of the bubble size distribution. However, whilst various theories and models have been proposed in the literature for the calculation of bubble breakup and coalescence rates, it must

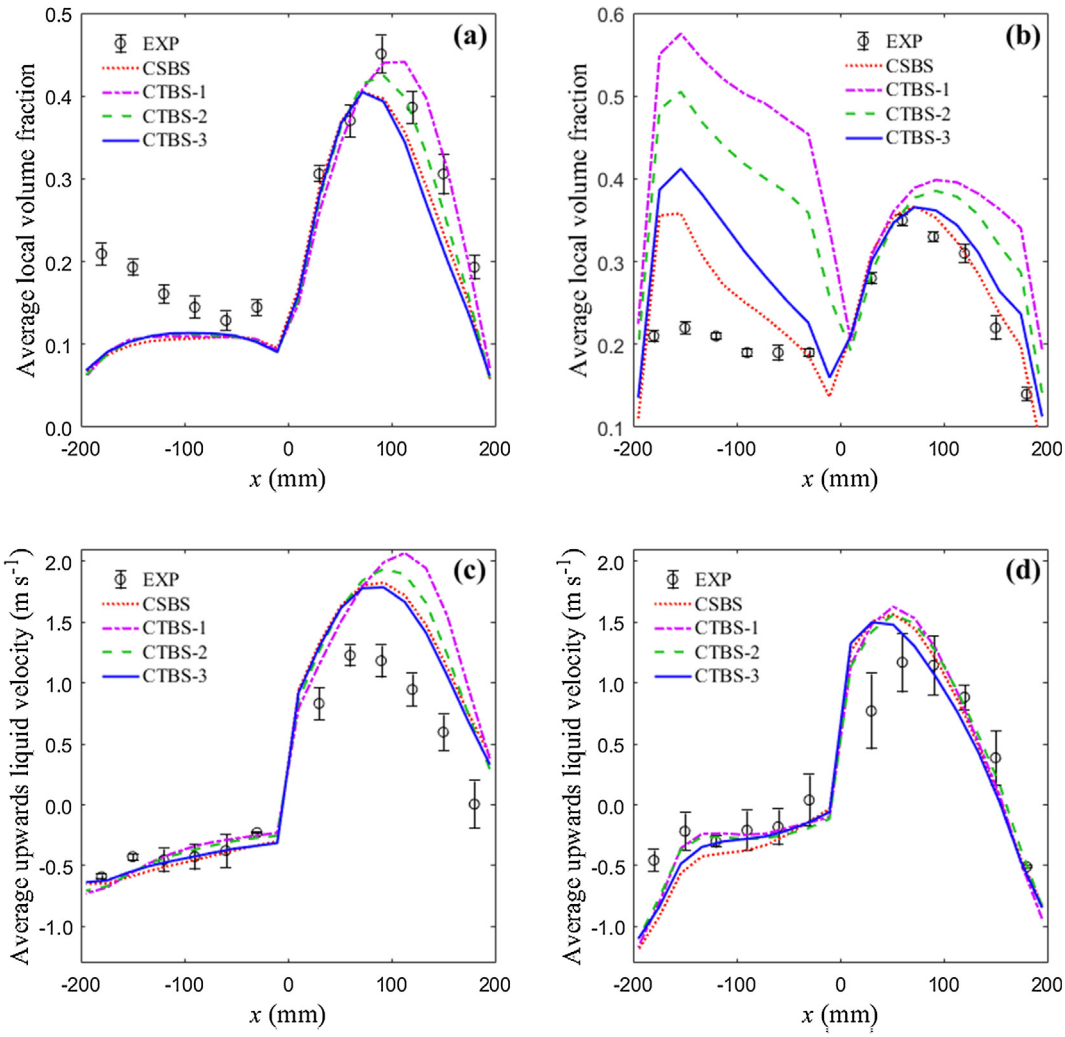

Fig. 8. Plots showing the local volume fraction and average upwards liquid velocity for the airlift column for a superficial gas velocity of 0.16 m s^{-1}. The top row shows hold-up profiles at heights of (a) 550 mm and (b) 1050 mm; while the bottom row shows profiles of the average upwards liquid velocity at (c) 550 mm and (d) 1050 mm above the base of the column. Results are shown for a constant, single bubble size (7 mm) model and constant, two bubble size models with 3 and 12 mm bubbles; the ratios of small to large bubbles are 80:20, 50:50 and 20:80 for models CTBS-1, CTBS-2 and CTBS-3, respectively.

be said that as yet there does not seem to be any agreement on what are the most appropriate options for either of these processes across the range of operating conditions employed in existing bubble column configurations. It is hoped that the experimental and simulated results presented in this study will promote future research in this area.

5. Conclusions

In the present numerical study, the impact of various bubble size models on CFD predictions has been systematically investigated for three bubble column configurations operating in the churn turbulent flow regime. The models considered employed a single bubble size (the CSBS and VSBS models), two bubble sizes (the CTBS model), as well as multiple bubble sizes (the homogeneous and inhomogeneous MUSIG models). Simulation results were compared with experimental measurements at two superficial gas velocities (0.16 and 0.25 m s^{-1}).

The CSBS model gave reasonable agreement with experimental data for the (symmetric) bubble column configuration, but less so for both the (asymmetric) half-sparger and airlift column configurations. The VSBS model gave similar predictions to the CSBS model but additionally its predicted mean bubble size was very much in line with experimental data. Thus, in addition to the simplicity and computational efficiency of the VSBS model, a key advantage is that the bubble size does not need to be specified as a model input parameter, which makes this option potentially beneficial in industrial circumstances where typically only limited bubble size data are available and experimental measurements can be challenging.

Surprisingly, neither the homogeneous nor the inhomogeneous MUSIG models provided improved predictions relative to the single bubble size models; a result attributed to the fact that these models over-predicted the amount of small bubbles. As previously noted, these results clearly highlight the need for improved breakup and coalescence closure descriptions as well as detailed experimental data for the validation of such models.

In conclusion, this study set itself a challenging brief, that is, to apply and evaluate different methodologies for modelling the BSD in large diameter, two-phase contactors; three bubble column configurations being considered, each one operated at the high superficial velocities typical of industrial scale applications. The authors are not aware of any similar study in the open literature. As a result, key areas for future development (both computational and experimental) have been identified, whilst the VSBS model was identified as a suitable predictive modelling option in situations where limited information is available on the likely bubble size distribution.

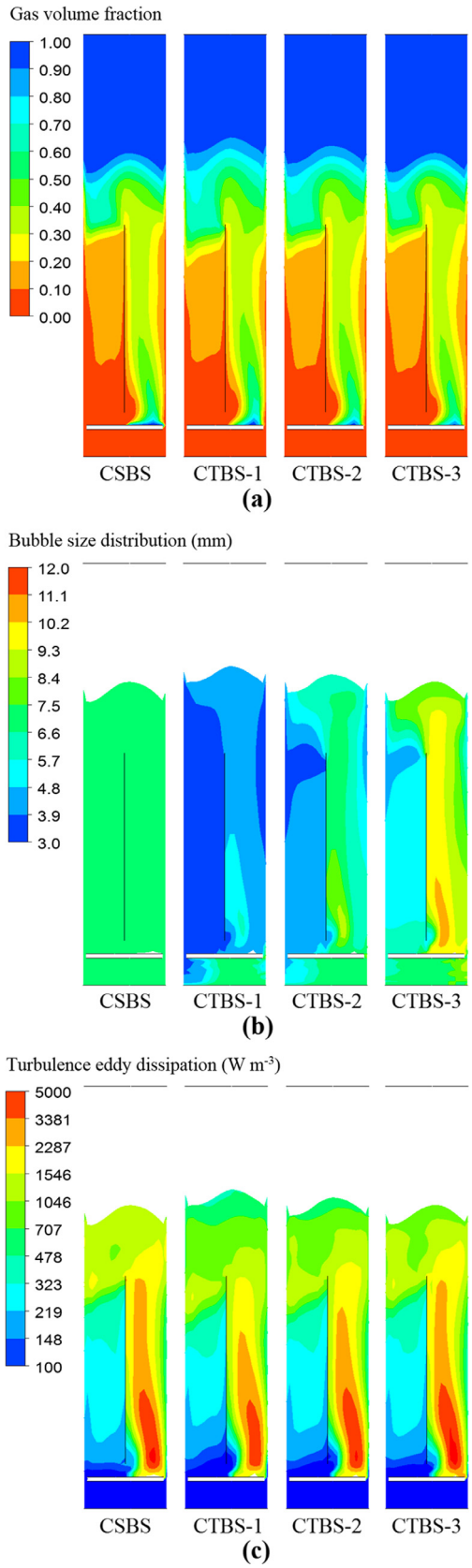

Fig. 9. Comparison of the predicted average (a) local gas volume fraction, (b) bubble size distribution and (c) liquid turbulence energy dissipation rate for the CTBS model variants for the airlift column configuration. The results are for a superficial gas velocity of 0.16 m s^{-1}.

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